

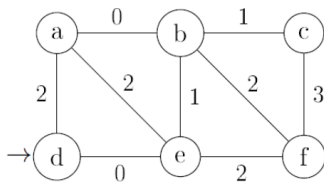
Algorithms and Data Structures II.

Endterm Test Sample

Note: A sample test was created with examples from the practice class material.

Task 1. (12 points)

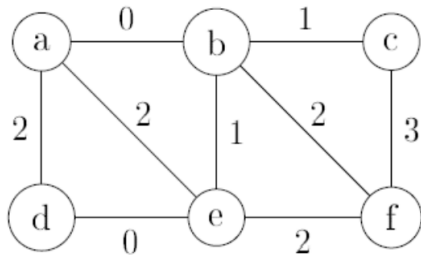
Illustrate the run of **Prim's** algorithm on the following graph with a source vertex **d**.
 Fill the table with the values of c (cost), *vertex into MST* (the newly added vertex), and $\pi(p)$ (parents) after every iteration of the algorithm.
 (Use the alphabetical or ascending order of the vertices.)
 Draw the minimum spanning tree after the algorithm terminates.



c values of nodes in Q						vertex into MST	changes of labels p					
a	b	c	d	e	f		a	b	c	d	e	f

Task 2. (12 points)

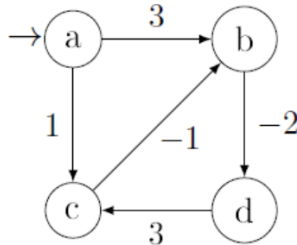
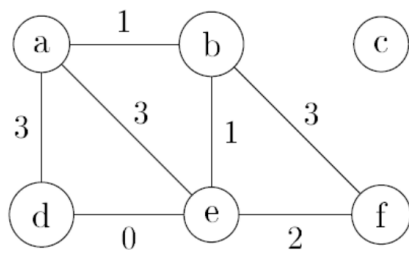
Illustrate the run of **Kruskal's** algorithm on the following graph.
 Draw the step-by-step transformation of the spanning forest.
 Fill the table with the values of *component*, *edge*, and *safe*.
 (Use the alphabetical or ascending order of the components formed by the contained vertices in order.)



Components	edge	used / safe?

Task 3. (12 points)

Illustrate the run of **Dijkstra's** algorithm on the following graph with a source vertex **a**.
 Fill the table with the values of d (distance/path cost), *expanded vertex* (newly added vertex), and $\pi(p)$ (parent).
 Draw the shortest-paths tree.
 (Use the alphabetical or ascending order of the vertices.)



Present your algorithm in structogram format.

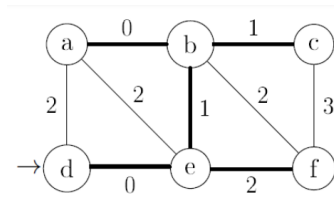
Algorithms and Data Structures II.

Endterm Test Sample Solutions

Note: A sample test was created with examples from the practice class material.

Task 1. (12 points)

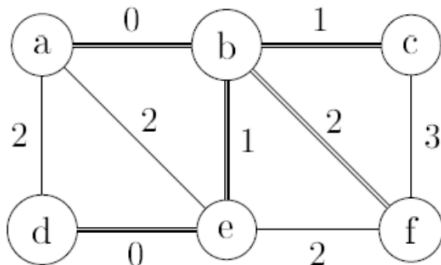
Illustrate the run of **Prim's** algorithm on the following graph with a source vertex 1. Fill the table with the values of c (cost), MST (the newly added vertex), and $\pi(p)$ (parents). Draw the minimum spanning tree. (Use the alphabetical or ascending order of the vertices.)



c values of nodes in Q						vertex into MST	changes of labels p					
a	b	c	d	e	f		a	b	c	d	e	f
∞	∞	∞	0	∞	∞	d	$\otimes$	$\otimes$	$\otimes$	$\otimes$	$\otimes$	$\otimes$
2	∞	∞		0	∞	d	d				d	
2	1	∞			2	e		e				e
0		1			2	b	b		b			
		1			2	a						
					2	c						
						f						
0	1	1	0	0	2	result	b	e	b	$\otimes$	d	e

Task 2. (12 points)

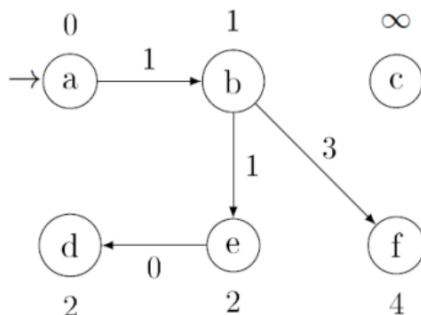
Illustrate the run of **Kruskal's** algorithm on the following graph. Fill the table with the values of *component*, *edge*, and *used/safe*. Draw the step-by-step transformation of the spanning forest. (Use the alphabetical or ascending order of the components formed by the contained vertices in order.)



Components	edge	used / safe?
a, b, c, d, e, f	(a,b)	+
ab, c, d, e, f	(d,e)	+
ab, c, de, f	(b,c)	+
abc, de, f	(b,e)	+
abcde, f	(a,d)	-
abcde, f	(a,e)	-
abcde, f	(b,f)	+
abcdef	-	-

Task 3. (12 points)

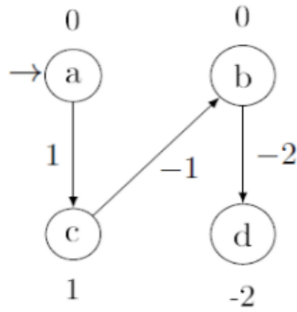
Illustrate the run of **Dijkstra's** algorithm on the following graph with a source vertex **a**. Fill the table with the values of d (distance/path cost), *expanded vertex* (newly added vertex), and $\pi(p)$ (parent). Draw the shortest-paths tree. (Use the alphabetical or ascending order of the vertices.)



d values of nodes in Q						expanded vertex : d	changes of π values					
a	b	c	d	e	f		a	b	c	d	e	f
0	∞	∞	∞	∞	∞	a : 0	$\otimes$	$\otimes$	$\otimes$	$\otimes$	$\otimes$	$\otimes$
	1	∞	3	3	∞	b : 1		a		a	a	
		∞	3	2	4	e : 2				e		
		∞	2		4	d : 2						
		∞				f : 4						
0	1	∞	2	2	4	result	$\otimes$	a	$\otimes$	e	b	b

Task 4. (12 points)

Illustrate the run of **Queue-based Bellman-Ford** algorithm on the following graph with a source vertex **a**. Fill the table with the values of d (distance/path cost), *expanded vertex* (newly added vertex), and $\pi(p)$ (parent). Draw the shortest-paths tree. (Use the alphabetical or ascending order of the vertices.)



changes of $d; e$				expanded vertex : $d; e$	Q : Queue	changes of π			
a	b	c	d			a	b	c	d
0; 0	∞	∞	∞		$\langle a \rangle$	$\otimes$	$\otimes$	$\otimes$	$\otimes$
	3; 1	1; 1		a: 0; 0	$\langle b, c \rangle$		a	a	
			1; 2	b: 3; 1	$\langle c, d \rangle$				b
	0; 2			c: 1; 1	$\langle d, b \rangle$		c		
				d: 1; 2	$\langle b \rangle$				
			-2; 3	b: 0; 2	$\langle d \rangle$				b
				d: -2; 3	$\langle \rangle$				
0	0	1	-2	final d and π values		$\otimes$	c	a	b